

UNIT -III

Classification: Classification problems, Evaluation of classification models, Bayesian methods, Logistic regression, neural networks, Support vector machines.

Clustering: Clustering methods, Partition methods, Hierarchical methods, Evaluation of clustering models

Classification :

1. **Definition:** Classification models are supervised learning methods used to predict the value of a **categorical target attribute** (like categories or labels).
2. **Difference from Regression:** Unlike **regression models**, which predict **numerical values**, classification deals with **categories**.
3. **How It Works:** These models learn from **past observations** where the target class is known and generate **rules** to predict the **class of future data**.
4. **Importance in Learning Theory:** Classification is important because it helps develop **algorithms** that learn from **past experiences**, similar to how the **human brain** learns.
5. **Real-Life Applications:**

- **Marketing:** Identifying **target customers**.
- **Fraud Detection:** Spotting **fraudulent activities**.
- **Image Recognition:** Identifying **objects in images**.
- **Medical Diagnosis:** Detecting **diseases early**.
- **Text Classification:** Organizing **documents and emails** (like spam detection).

6. **Chapter Overview:** This chapter explains:

- **Characteristics of classification problems.**
- **How to evaluate and compare models.**
- **Major classification methods:**
 - **Classification Trees**
 - **Bayesian Methods**
 - **Neural Networks**
 - **Logistic Regression**
 - **Support Vector Machines (SVM)**

Classification Problems :

1. Dataset Description:

- Dataset **D** has **m observations** (examples/instances) and **n explanatory attributes** (predictive variables).
- Explanatory attributes can be **categorical** or **numerical**.
- The **target attribute** (class/label) is **categorical** with a **finite number of values**.

2. Types of Classification:

- **Binary Classification:** Two classes only.
- **Multiclass Classification:** More than two classes.

3. Purpose of Classification Models:

- Identify **relationships** between explanatory variables and target class.
- Create **classification rules** to predict the **class of future examples**.

4. Model Characteristics:

- **Rules format** depends on the **type of model used**.
- Most models are **nonparametric**, meaning they **don't assume any specific data distribution**.

5. Components of a Classification Problem:

- **Generator:** Extracts random vectors of examples based on an **unknown probability distribution**.
- **Supervisor:** Assigns a **target class** to each example using an **unknown conditional distribution**.
- **Algorithm:** Also called a **classifier**, chooses a *function* $(f)^*$ to **minimize loss** and improve accuracy.

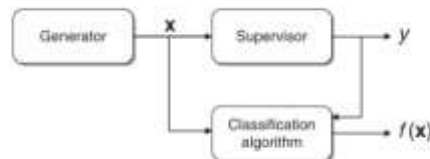


Figure 10.1 Probabilistic structure of the learning process for classification

6. Interpretation vs. Prediction:

- **Simple models:** Easy to interpret but may have **lower accuracy**.
- **Advanced models:** Harder to interpret but **higher prediction accuracy**.

7. Training and Testing:

- **Training Phase:**

1. Use a **subset (T)** of the dataset **D** (training set) to **derive classification rules**.
- **Test Phase:**
 1. Use the **remaining data ($V = D - T$)** to **test model accuracy** by comparing **predicted vs. actual class**.
 2. **Training and test sets must be disjoint** to avoid overestimating accuracy.
8. **Prediction Phase:**
 - Apply the **classification rules** to **new data** for **future predictions**.
9. **Tuning Set:**
 - A **part of the training data** used to **fine-tune model parameters** for **optimal performance**.

Taxonomy of classification models :

Types of Classification Models:

- **Heuristic Models:**
 - Use simple and intuitive algorithms.
 - Examples: Nearest neighbor methods (based on distance) and classification trees (divide-and-conquer).
 - **Example:** Imagine you want to classify fruits as "Apple" or "Orange" based on their color and weight. Nearest neighbor method would check which nearby fruit (in terms of color and weight) it looks most like and assign the same label.
- **Separation Models:**
 - Divide the attribute space into disjoint regions based on target class.
 - Misclassified points are handled with a loss function and optimization.

- Examples: Discriminant analysis, perceptron methods, neural networks, support vector machines, some classification trees.
- **Example:** Think of drawing a line on a paper to separate red dots (Apples) from orange dots (Oranges). Any new dot on one side of the line is an Apple, and on the other side is an Orange.
- **Regression Models:**
 - Assume a functional form of conditional probabilities.
 - Used for binary classification through logistic regression (extension of linear regression).
 - Example: If we know height and weight of people, and want to predict if a person is "Underweight" or "Overweight", logistic regression creates a curve to separate the two categories based on the numbers.
- **Probabilistic Models:**
 - Use conditional probabilities of observations given the target class.
 - Apply Bayes' theorem to calculate posterior probabilities.
 - Examples: Naive Bayes classifiers, Bayesian networks.
 - **Example:** If 80% of the time a round, red fruit is an Apple and 20% it's a Tomato, the model will predict Apple because of the higher probability. Naive Bayes works like this by calculating these chances.

Score Function:

- Associates each observation with a real number (probability estimate).
- In separation models: Distance from class boundary.
- In classification trees: Density of observations with the same target class.
- **Example:** Suppose we score a student's performance out of 100. Higher the score, higher the chance they pass. Similarly, a

classification model gives a number to each observation to show how likely it belongs to a certain class.

Classification Rule:

- Derived from the score function to predict the target class.
- For binary classification: Uses the sign of the score function to determine class $\{-1,1\}$.
- **Example:** If we classify emails as "Spam" or "Not Spam", and the model gives a spam score. If the score is above 0.5, we classify it as spam. If it's below 0.5, it's not spam.

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Evaluation of Classification Model :

1. Developing Alternative Models:

- Use different methods like classification trees, neural networks, Bayesian techniques, or support vector machines.
- Modify parameter values to improve prediction accuracy.

2. Evaluation Criteria for Classification Methods:

- **Accuracy:** Measures how well the model predicts the target class for future observations.
- Formula for accuracy: Proportion of correctly classified observations in the test set.
- Formula for error rate: Proportion of misclassified observations in the test set.
- **Speed:** Some methods are faster and better for larger datasets.

- **Robustness:** A robust model maintains accuracy across different training and test sets and handles missing data and outliers well.
- **Scalability:** Ability to learn from large datasets efficiently.
- **Interpretability:** Simple, easy-to-understand rules for domain experts.

3. Holdout Method:

- Split data into two subsets: training (T) and test (V) sets.
- Test set accuracy depends on the selected subset and may overestimate or underestimate the model's accuracy.

4. Repeated Random Sampling:

- Repeat the holdout method multiple times.
- Each time, randomly select a new training set and calculate accuracy.
- Overall accuracy is the average of individual accuracies.
- More repetitions provide a more reliable accuracy estimate.

5. Cross-Validation:

- Common choice: Tenfold cross-validation (split dataset into 10 subsets).
- Leave-one-out method: Each test set has one observation, maximizing training data for each iteration.

6. Stratified Random Sampling:

- Ensures each subset has the same proportion of target class observations as the entire dataset.

7. Confusion Matrix:

- For binary classification, shows true and false predictions for both classes.
- Metrics:
 - **True Negatives (p):** Correctly predicted negative examples.
 - **False Negatives (u):** Incorrectly predicted positive examples.

- **False Positives (q):** Incorrectly predicted negative examples.
- **True Positives (v):** Correctly predicted positive examples.

8. Performance Metrics:

- **False Negative Rate:** $u / (u + v)$
- **False Positive Rate:** $q / (p + q)$
- **True Positive Rate (Recall):** $v / (u + v)$
- **Precision:** $v / (q + v)$
- **Geometric Mean:** $tp \times prc$ or $tp \times tn$
- **F-measure:** Weighted balance between precision and recall.

9. ROC Curve:

- Visual representation of model performance by plotting true positive rate vs. false positive rate.

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Bayesian methods :

1. Bayesian Methods Overview:

- Belong to probabilistic classification models.
- Calculate posterior probability $P(y|x)$ using Bayes' theorem.
- Requires prior probability $P(y)$ and class conditional probability $P(x|y)$.

2. Bayesian Classifiers:

- Estimate probability $P(x|y)$ for classification.
- Learning phase involves analyzing training set observations.

3. Bayes' Theorem:

- Formula: $P(y|x) = [P(x|y)P(y)] / P(x)$
- Maximum a posteriori hypothesis (MAP) assigns class with highest $P(y|x)$.
- 4. **Estimating Prior Probability:**
 - $P(y=vh) = mh / m$ (Frequency of class value in dataset).
- 5. **Computational Challenge:**
 - Estimating $P(x|y)$ requires a huge number of observations for accuracy.
- 6. **Simplifying Assumptions:**
 - Leads to Naive Bayes classifiers and Bayesian networks.
- 7. **Naive Bayesian Classifier:**
 - Assumes conditional independence of attributes given target class.
 - Formula: $P(x|y) = P(x_1|y) * P(x_2|y) * ... * P(x_n|y)$
- 8. **Estimating Probabilities:**
 - **Categorical/Discrete Attributes:**
 - $P(x_j|y) = sjhk / mh$ (Frequency of attribute value in class).
 - **Numerical Attributes:**
 - Assumes distribution like Gaussian density.
- 9. **Accuracy:**
 - Despite simplifying assumptions, achieves accuracy comparable to complex methods.
- 10. **Example Usage:**
 - Relative frequencies of attributes (like area, numin, timein) given target class.
 - Calculates probabilities for class prediction.

Logistic regression :

1. What is Logistic Regression in BI and Data Analytics?

- **Logistic Regression** is a **predictive analytics technique** used to **classify outcomes into two categories** — like **Yes/No, Success/Failure, 1/0**.
- In **Business Intelligence (BI)** and **Data Analytics**, it helps businesses make **data-driven decisions** by predicting future outcomes based on past data.

2. Why Use Logistic Regression in BI?

- **Customer Behavior Prediction:** Will a customer buy a product or not?
- **Churn Analysis:** Will a customer leave the service or stay?
- **Risk Analysis:** Will a loan default happen or not?
- **Marketing Campaigns:** Will the campaign be successful or not?

3. How Does Logistic Regression Work?

It **predicts the probability** of an event happening.

- Uses the **Sigmoid Function** (S-shaped curve) to **convert predictions to probabilities** between **0 and 1**.

- Formula:

$$P(y = 1|x) = \frac{e^{wx}}{1 + e^{wx}}$$

Where:

- $P(y = 1|x)$ = Probability of event happening (like a customer buying)
- w = Weights (importance of factors)
- x = Input variables (like customer age, income)
- e = Euler's number (~2.718)

4. Business Example: Customer Purchase Prediction

Problem: Predict if a customer will **buy a product** based on their data.

Data:

- Age
- Income
- Past Purchases

Model Equation:

$$wx = -5 + 0.03(\text{Age}) + 0.01(\text{Income})$$

For a customer:

- Age = 30
- Income = 50,000

Calculate:

$$wx = -5 + 0.03(30) + 0.01(50000) = -5 + 0.9 + 500 = 495.9$$

Apply Sigmoid Function:

$$P(\text{Buy} = 1) = \frac{1}{1 + e^{-495.9}} \approx 1$$

Interpretation: The customer has **almost a 100% chance** of buying the product.

5. Why Logistic Regression is Useful in BI and Analytics

- **Easy to Interpret:** Gives clear probability scores for decision-making.
- **Efficient for Binary Decisions:** Works well for **Yes/No** type predictions.
- **Helps Optimize Strategies:** Businesses can **target high-probability customers**.

6. Challenges and Considerations

- **Not Good for Complex Relationships:** Works best when **inputs and outputs are linearly related**.
 - **Requires Clean Data:** Missing or inconsistent data can **reduce accuracy**.
 - **Assumes Independence:** Assumes that **input variables are not strongly correlated**.
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1. What is a Neural Network?

- **Neural Networks (NN)** are **AI-based algorithms** inspired by how the **human brain works**.
- They help computers **recognize patterns, learn from data, and make decisions**.
- In **Business Intelligence (BI)** and **Data Analytics**, they are used for **advanced predictions, classification, and decision-making**.

2. Why Use Neural Networks in BI and Data Analytics?

- **Handle Complex Data:** Works well with **large and complicated datasets**.
- **Pattern Recognition:** Finds **hidden trends and relationships** in data.
- **Automation:** Reduces the need for **manual analysis**.
- **High Accuracy:** Provides **better predictions** compared to traditional models like **Logistic Regression**.

3. Structure of a Neural Network

Neural networks are made up of **layers of nodes (neurons)**:

a) Input Layer:

- Takes **raw data** like **customer age, income, purchase history**, etc.

b) Hidden Layers:

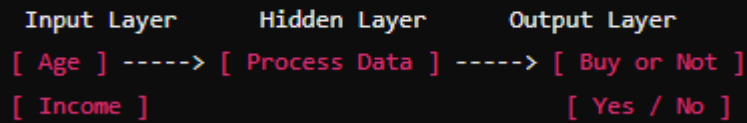
- **Processes the input and extracts patterns**.
- Applies **mathematical operations** (like weights and activation functions).

c) Output Layer:

- **Gives the final prediction** — like whether a customer will buy or not.

Example Structure:

CSS



4. How Neural Networks Work

- **Step 1: Input Data:** Feed **customer data** into the network.
- **Step 2: Weights:** Assign **importance to each input** (like how much income impacts buying).
- **Step 3: Activation Function:** Decides whether to **activate a neuron** based on weighted input.
- **Step 4: Output:** Predicts the **final result** — like “**Will the customer buy?**”

5. Business Example: Predicting Product Purchase

Problem: Will a customer buy a product based on their data?

Data:

- **Age:** 28
- **Income:** ₹60,000
- **Past Purchases:** 5

Neural Network Flow:

Input Layer		Hidden Layer		Output Layer
[28]	----->	[Weights + Bias]	----->	[1 = Buy]
[60000]	----->	[Activation Fn]	----->	[0 = No Buy]
[5]	----->		----->	[Probability]

Prediction:

- If output is **close to 1** → **Likely to buy**
- If output is **close to 0** → **Unlikely to buy**

6. Advantages of Neural Networks in BI

- **Handles Large Data:** Works well with **big data** and **complex datasets**.
- **High Accuracy:** Better than traditional models for **difficult predictions**.
- **Automated Learning:** **Learns and improves** over time with more data.

7. Challenges of Neural Networks

- **Needs Lots of Data:** Works best with **large amounts of quality data**.
- **Computationally Heavy:** Requires **high processing power**.
- **Difficult to Interpret:** Unlike simple models, it's **hard to understand why** decisions are made.

8. Real-Life Applications in BI and Analytics:

- **Customer Churn Prediction:** Will a customer leave the service?
- **Sales Forecasting:** Predict **future product demand**.
- **Fraud Detection:** Identify **suspicious transactions**.
- **Sentiment Analysis:** Understand **customer feedback** from reviews.

Support vector machines:

1. What is Support Vector Machine (SVM)?

- **SVM is a machine learning algorithm** used for **classification** and **regression** tasks.
- It's most commonly used to **classify data into two or more categories**.
- SVM works by **drawing a line (or hyperplane)** that **best separates the data** into different groups.

2. Why Use SVM in BI and Data Analytics?

- **High Accuracy:** Works well even with **complex data**.
- **Effective for Classification:** Great at **separating different types of data**.
- **Handles High-Dimensional Data:** Works with **many features** (like customer age, income, behavior, etc.).
- **Outlier Resistant:** SVM focuses on the **most important data points** (called **support vectors**) to create boundaries.

3. How Does SVM Work?

- **Step 1: Input Data:** Data has **two or more categories** (like "Buy" or "Not Buy").
- **Step 2: Draw a Line:** SVM finds the **best possible line (or hyperplane)** that separates the two categories.
- **Step 3: Maximize Margin:** SVM **maximizes the distance** between the line and the **closest data points from both groups**.
- **Step 4: Classify New Data:** When new data comes in, SVM checks **which side of the line it falls on** and **classifies it**.

4. Key Terms in SVM:

- **Hyperplane:** The line (in 2D) or plane (in 3D+) that separates data categories.
- **Support Vectors:** The data points closest to the hyperplane — they define the boundary.
- **Margin:** The distance between the hyperplane and support vectors — wider margins = better classification.

5. Business Example: Classifying Customers

Problem: Classify customers as “**High Spenders**” or “**Low Spenders**” based on their behavior.

Data:

- **Age:** 28
- **Income:** ₹60,000
- **Number of Purchases:** 5

SVM Flow :

```
High Spenders      |      Low Spenders
[ Age: 30 ]         |      [ Age: 25 ]
[ Income: ₹80K ]    |      [ Income: ₹40K ]
[ Purchases: 10 ]   |      [ Purchases: 3 ]

Hyperplane: The best line that separates high and low spenders.
```

New Customer:

- **Age:** 28

- **Income:** ₹60,000
- **Purchases:** 5

SVM Decision:

- Falls closer to “**High Spenders**” → Classified as **High Spender**.

6. Advantages of SVM in BI and Analytics:

- **Works Well with Complex Data:** Great for **non-linear relationships** using **kernel tricks**.
- **High Accuracy:** Even with **small datasets**.
- **Robust to Outliers:** Focuses only on the **most important data points** (support vectors).
- **Effective in High Dimensions:** Works well even with **many features** (like customer data).

7. Challenges of SVM:

- **Computationally Intensive:** Slow with **very large datasets**.
- **Hard to Tune:** Requires choosing the **right kernel** and **parameters** for best results.
- **Difficult Interpretation:** Hard to understand the model’s decisions compared to simpler models.

8. Real-Life Applications in BI and Analytics:

- **Customer Segmentation:** Grouping customers based on behavior.
- **Fraud Detection:** Identifying **suspicious transactions**.
- **Sales Forecasting:** Predicting **future sales trends**.

- **Sentiment Analysis:** Classifying **customer feedback** as positive, neutral, or negative.

Clustering :

1. What is Clustering?

- **Clustering is a technique used to group similar data points together.**
- **It finds patterns in data by grouping similar items into clusters.**
- **Objects in the same cluster are similar, and objects in different clusters are different.**

2. Why Use Clustering in BI and Data Analytics?

- **Discover Hidden Patterns:** Helps **find relationships** between data that aren't obvious.
- **Customer Segmentation:** Groups customers by **behavior, preferences, or spending patterns.**
- **Market Analysis:** Identifies **distinct groups in a market** for targeted campaigns.
- **Anomaly Detection:** Detects **unusual behavior or outliers** (like fraudulent transactions).

3. How Does Clustering Work?

- **Step 1: Collect Data:** Data with **different attributes** (like age, income, purchase history).
- **Step 2: Choose Number of Clusters:** Decide how many **groups or clusters** you want.
- **Step 3: Group Similar Data:** Algorithm **groups data points** based on **similar features**.
- **Step 4: Analyze Clusters:** Each **cluster** shows a **different type of pattern or behavior**.

4. Types of Clustering Algorithms:

- **K-Means Clustering:**
 - **Most common technique.**
 - Divides data into **K clusters** where **K is the number of groups** you choose.
 - **Example:** Grouping customers into **3 categories: Low, Medium, and High spenders**.
- **Hierarchical Clustering:**
 - **Creates a tree of clusters (dendrogram).**
 - **No need to pre-define the number of clusters.**
 - **Example:** Grouping **employees by skills and experience**.
- **DBSCAN (Density-Based Clustering):**
 - Groups **dense areas of data** and **identifies outliers**.
 - **Example:** Detecting **fraudulent transactions** in a dataset.

5. Business Example: Customer Segmentation

- **Problem:** A retail store wants to **group customers based on their behavior**.
- **Data:**

Customer ID	Age	Monthly Spend (₹)	Visits Per Month
1	25	10,000	5
2	45	50,000	12
3	30	20,000	7
4	50	70,000	15

Clustering Results:

- **Cluster 1:** Young customers, **low spending**, few visits.
- **Cluster 2:** Middle-aged customers, **medium spending**, moderate visits.
- **Cluster 3:** Older customers, **high spending**, frequent visits.

Insights:

- **Target Cluster 3 with premium offers.**
- **Offer discounts to Cluster 1 to increase spending.**

6. Advantages of Clustering in BI and Analytics:

- **Unsupervised Learning:** No need for labeled data; groups form based on patterns.
- **Discover Insights:** Reveals **hidden trends and relationships** in data.
- **Better Decision Making:** Helps in **targeted marketing, personalized services**, and **resource allocation**.
- **Easy to Visualize:** Clusters can often be **shown in charts or graphs** for better understanding

7. Challenges of Clustering:

- **Choosing Number of Clusters:** Deciding the **right number of groups** can be tricky.
- **Scalability Issues:** Slow performance with large datasets.
- **Sensitive to Outliers:** Outliers can affect cluster formation.
- **Interpretation:** Understanding **why clusters formed** can sometimes be difficult.

8. Real-Life Applications in BI and Analytics:

- **Customer Segmentation:** Grouping customers by **buying behavior**.
 - **Product Recommendation:** Suggesting products based on **similar customer choices**.
 - **Fraud Detection:** Identifying **unusual transactions** as outliers.
 - **Market Research:** Understanding **different customer needs**.
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Clustering Methods :

A) Partitioning Methods: (K Means)

- **What it does:** Divides data into a **fixed number of clusters**.
- **How it works:**
 1. Choose the **number of clusters (K)**.
 2. Assign **each data point** to a cluster.
 3. Adjust clusters to **minimize differences within a cluster** and **maximize differences between clusters**.
- **Popular Algorithm: K-Means Clustering**
 - **How K-Means works:**
 1. Choose **K (number of clusters)**.
 2. Pick **K random points** as "**centroids**" (cluster centers).
 3. Assign **each data point** to the **nearest centroid**.
 4. **Recalculate centroids** by averaging points in each cluster.
 5. **Repeat steps 3-4 until clusters stabilize**.

- **Example:**

Customer ID	Age	Monthly Spend (₹)	Visits Per Month
1	25	10,000	5
2	45	50,000	12
3	30	20,000	7
4	50	70,000	15

- **Clustering Results:**

- **Cluster 1:** Young customers, **low spending**, few visits.
- **Cluster 2:** Middle-aged customers, **medium spending**, moderate visits.
- **Cluster 3:** Older customers, **high spending**, frequent visits.

- **Advantages:**

- **Simple and fast** for small datasets.
- **Easy to interpret** results.

- **Disadvantages:**

- **Need to decide the number of clusters (K)** in advance.
 - **Sensitive to outliers.**
 - **Doesn't work well with non-spherical clusters.**
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B) Hierarchical Methods:

- **What it does:** Builds a **tree of clusters** (called a dendrogram).

- **How it works:**

- Starts with **each data point as its own cluster**.
- **Gradually merges** the most similar clusters until **only one cluster remains**.
- The **tree structure** shows how clusters are related.

- **Types:**

1. **Agglomerative (Bottom-Up):**

- Starts with **individual points** and **combines them into larger clusters**.

2. **Divisive (Top-Down):**

- Starts with one big cluster and splits it into smaller clusters.

- **Example:**

Product ID	Price (₹)	Popularity Score
1	1000	80
2	5000	60
3	2000	90

Dendrogram Output:

- Cluster 1: Affordable, high popularity
- Cluster 2: Expensive, moderate popularity
- **Advantages:**
 - **No need to specify the number of clusters beforehand.**
 - **Captures hierarchy and relationships** between data.
- **Disadvantages:**
 - **Computationally heavy** for large datasets.
 - **Difficult to interpret dendrograms** with too many data points.

Evaluation of clustering models :

We're discussing how to **evaluate clustering models** in **unsupervised learning**.
Let's understand some basics first:

- **Supervised Learning:**

- We have **input data and target labels** (like "Spam" or "Not Spam" for emails).
- **Evaluation is easy** because we can compare predictions to actual labels using **accuracy, precision, recall**, etc.
- **Unsupervised Learning (like Clustering):**
 - **No target labels**, just **raw data**.
 - We group similar items together without knowing their categories beforehand.
 - **Evaluation is harder** because we can't directly compare to actual categories.

2. What Is Clustering Evaluation? Why Is It Important?

Since clustering doesn't have labeled data, we use **different measures** to check:

- Are **similar items grouped together**?
- Are **different items well separated**?
- How **good and meaningful** are the clusters?

Real-Life Example:

Imagine you're working in a **retail store** and want to **group customers** based on their shopping behavior. You don't know their "type" — you just have data like:

Customer Age Monthly Spend (₹) Visits/Month

A	25	10,000	5
B	45	50,000	12
C	30	20,000	7

We use **clustering** to group customers, but how do we **know if the groups make sense**? That's where **clustering evaluation** comes in.

3. Steps to Evaluate Clustering Models:

Step 1: Check for Actual Patterns in Data

- We first check if the clusters **really reflect meaningful patterns** in the data.
- How? By applying **different clustering algorithms** and **comparing results**.

- If clusters are similar across methods, it shows **stability and reliability**.

Example: Apply **K-Means**, **Hierarchical Clustering**, and **DBSCAN** — if they show **similar groupings**, the clusters are **likely valid**.

Step 2: Apply Performance Indicators

We use **mathematical measures** to check how **good or bad the clusters are**:

A) Cohesion (Within-Cluster Similarity)

- Measures **how similar the points are within the same cluster**.
- **Lower cohesion = better clustering** (points are closer together).

Formula:

$$\text{Cohesion}(C_h) = \sum \text{dist}(x_i, x_k)$$

Where:

- x_i and x_k are points in the same cluster C_h .
- **dist()** is the distance between the points (like **Euclidean distance**).

Overall Cohesion for All Clusters:

$$\text{Coh}(C) = \sum \text{Coh}(C_h)$$

Example:

Cluster of customer spending:

- ₹10,000, ₹12,000, ₹11,500 → **Cohesion is low** → Customers are similar → **Good clustering**
 - ₹5,000, ₹50,000, ₹1,00,000 → **Cohesion is high** → Customers are very different → **Bad clustering**
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B) Separation (Between-Cluster Difference)

- Measures **how far apart different clusters are**.
- **Higher separation = better clustering** (clusters are more distinct).

Formula:

$$\text{Separation}(C_h, C_f) = \sum \text{dist}(x_i, x_k)$$

Where:

- x_i is from cluster C_h and x_k is from cluster C_f .

Overall Separation:

$$\text{Sep}(C) = \sum \text{Sep}(C_h, C_f)$$

Example:

- Cluster 1: Budget customers (₹1,000–₹5,000)
- Cluster 2: Premium customers (₹50,000–₹70,000)
- **High separation** because groups are very different → **Good clustering**

C) Silhouette Coefficient (Combination of Cohesion and Separation)

- Measures **how well each point fits into its cluster** compared to other clusters.
- Value ranges from **-1 to +1**:
 - **+1**: Perfectly assigned → **Good clustering**
 - **0**: On the **boundary between clusters**
 - **-1**: **Wrongly assigned** to the wrong cluster → **Bad clustering**

Steps to Calculate:

1. **Find Cohesion u_{iu}** : Average distance of point x_{ix} from **other points in its cluster**.
2. **Find Separation v_{iv}** : Average distance of point x_{ix} from **points in the nearest different cluster**.
3. **Silhouette Coefficient**:

$$s_i = \frac{v_i - u_i}{\max(u_i, v_i)}$$

Overall Silhouette Score:

- **Average silhouette coefficient** across all data points.

Example:

- Customer spending ₹12,000 is **very close to others in its cluster** → **Silhouette Score near +1.**
- Customer spending ₹15,000 is **equally close to two clusters** → **Score near 0.**
- Customer spending ₹50,000 is **closer to the wrong cluster** → **Negative score.**

4. Visualizing Clustering Quality (Silhouette Diagrams)

- **Silhouette diagrams** help visualize **how well data points fit into clusters.**
- **Vertical axis:** Data points, grouped by cluster.
- **Horizontal axis:** Silhouette coefficient value.
- **Wide, consistent bars:** **Good clustering.**
- **Narrow, mixed bars:** **Poor clustering.**

Example:

In a **customer segmentation model**, if most customers in **Cluster 1** have a silhouette score of **0.8+**, it's a **well-defined cluster**.

5. Comparing Different Clustering Techniques

It's **important to try different methods** like:

- **K-Means Clustering**
- **Hierarchical Clustering (Agglomerative/Divisive)**
- **Partitioning Around Medoids (PAM)**

Why?

- **If the clusters remain similar**, it shows they **reflect actual patterns in the data.**

- If they differ a lot, the clusters might be **unstable or not meaningful**